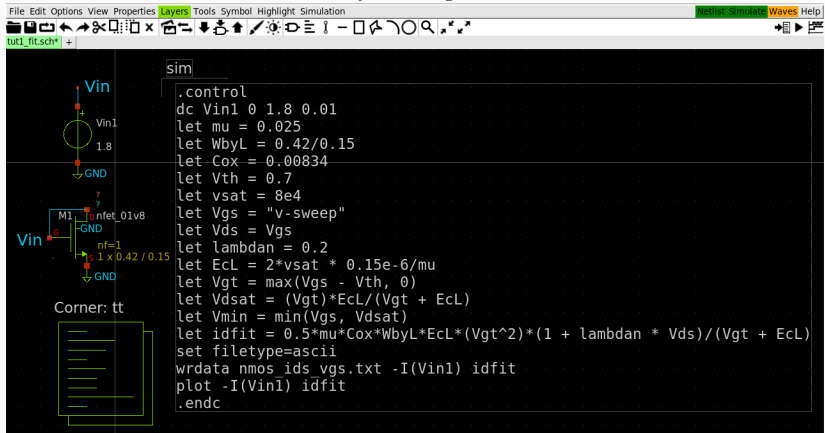


EE5311: Digital IC Design

Tutorial 2

Experiment 1(a) - xschem I_{DS} , $I_{DS,fit}$ vs V_{GS}

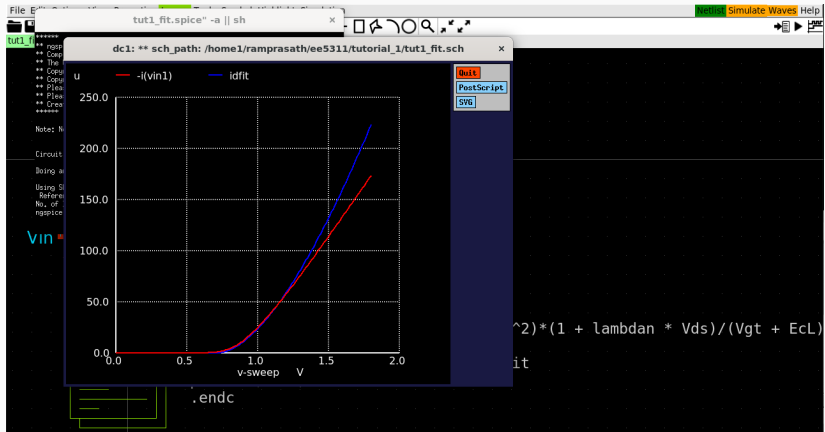
Plot and save the simulated/analytical expressions:



The screenshot displays the xschem simulation environment. On the left, a circuit diagram is shown with a voltage source V_{in1} (1.8V) connected to the gate of an nMOS transistor (M1). The gate is also connected to a node labeled V_{in} . The source is connected to ground (GND). The drain is connected to a node labeled V_{in} and a node labeled $01v8$. The transistor parameters are $n_f=1$ and $s=1 \times 0.42 / 0.15$. The corner is set to tt . On the right, a control script is shown in a text editor window titled `sim`. The script defines parameters for the simulation and plots the drain current I_{DS} and the fitted drain current $I_{DS,fit}$ versus the gate voltage V_{GS} .

```
.control
dc Vin1 0 1.8 0.01
let mu = 0.025
let WbyL = 0.42/0.15
let Cox = 0.00834
let Vth = 0.7
let vsat = 8e4
let Vgs = "v-sweep"
let Vds = Vgs
let lambdan = 0.2
let EcL = 2*vsat * 0.15e-6/mu
let Vgt = max(Vgs - Vth, 0)
let Vdsat = (Vgt)*EcL/(Vgt + EcL)
let Vmin = min(Vgs, Vdsat)
let idfit = 0.5*mu*Cox*WbyL*EcL*(Vgt^2)*(1 + lambdan * Vds)/(Vgt + EcL)
set filetype=ascii
wrdata nmos ids vgs.txt -I(Vin1) idfit
plot -I(Vin1) idfit
.endc
```

Experiment 1(a) - xschem I_{DS} , $I_{DS,fit}$ vs V_{GS}



Experiment 1(b) - I_{DS} vs V_{DS} for different V_{GS}

Plot I_{DS} vs V_{DS} for a unit-sized nMOS $\left(\frac{W}{L} = \frac{0.42\mu\text{m}}{0.15\mu\text{m}}\right)$ with $V_{GS} \in \{0.6V, 1.0V, 1.4V, 1.8V\}$

The screenshot shows the Xschem simulation environment with a circuit diagram and a SPICE netlist. The circuit diagram on the left includes two voltage sources, Vdd1 and Vin1, both set to 1.8V. Vdd1 is connected to the drain of an nMOS transistor (M1, nfet_01v8). The gate of M1 is connected to Vin1. The source of M1 is connected to ground. A corner analysis block is shown below the circuit diagram, labeled "Corner: tt". The SPICE netlist on the right is as follows:

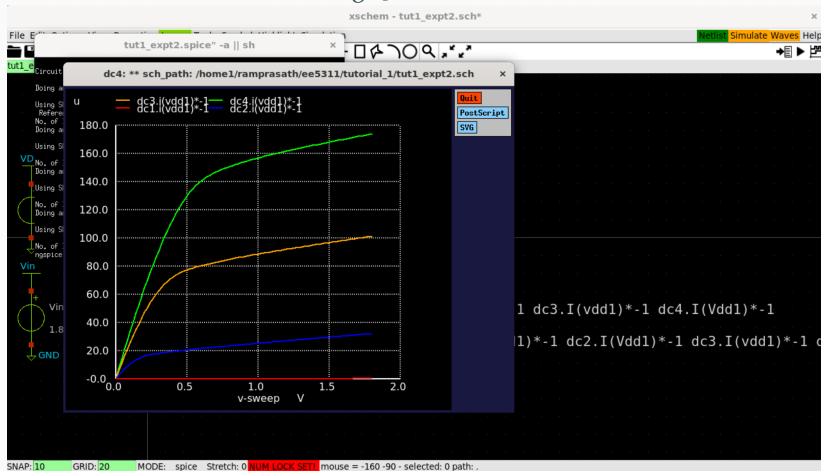
```
sim
.control
let Vgs = 0.6
repeat 4
alter Vin1 $&Vgs
dc Vdd1 0 1.8 0.02
let Vgs = Vgs + 0.4
end
plot dc1.I(Vdd1)*-1 dc2.I(Vdd1)*-1 dc3.I(vdd1)*-1 dc4.I(Vdd1)*-1
set filetype=ascii
wrdata nmos_ids_vds.txt dc1.I(Vdd1)*-1 dc2.I(Vdd1)*-1 dc3.I(Vdd1)*-1 dc4.I(Vdd1)*-1
.endc
```

The status bar at the bottom indicates: SNAP: 10 GRID: 20 MODE: spice Stretch: 0 NUM LOCK SET mouse = -160 80 - selected: 0 path: .

Experiment 1(b) - I_{DS} vs V_{DS} for different V_{GS}

The simulated I_{DS} vs V_{DS} data should be in the file: `nmos_ids_vds.txt`.

View the contents of the file using: `gedit nmos_ids_vds.txt`



Experiment 1(c)

Simulate the I_{DS} vs V_{DS} for $V_{GS} = V_{DD}$ for transistors of different W/L keeping W/L ratio constant while increasing the width.

The screenshot shows the xschem - tut1_1.sch window. The circuit diagram on the left includes a VDD source labeled Vdd1 with a value of 1.8, connected to the gate of an nmosfet_01v8 transistor (M1). The transistor's source is connected to GND, and its drain is connected to a node labeled 'G'. A voltage source Vin is connected to the gate of M1. The transistor's width is defined as $1 \times \text{Width} / \{\text{Length}\}$. The simulation script on the right is as follows:

```
sim
.param Width = 0.42
.param Length = 0.15
.dc Vdd1 0 1.8 0.01
.control
  let index = 1
  set cache = ''
  while index <= 10
    let newW = index * 0.42
    let newL = index * 0.15
    alterparam Width = $&newW
    alterparam Length = $&newL
    reset
    run
    set cache = ( $cache dc{$&index}.i(VDD1)*-1
    let index = index + 1
  end
  plot $cache
.endc
```

Below the circuit diagram, there is a corner analysis table for 'Corner: tt' with 10 rows and 2 columns, all of which are empty.

At the bottom of the window, the status bar shows: SNAP: 10, GRID: 20, MODE: spice, Stretch: 0, NUM LOCK SET, mouse = -80 210 - selected: 0 path: .

Experiment 1(c) – simulation output

